

ECE368: Probabilistic Reasoning

Lab 1: Classification with Multinomial and Gaussian Models

You can complete this lab in a group of two. Please provide the name and student number of both members.

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Name:	Student Number:

You should hand in: 1) A scanned .pdf version of this sheet with your answers (file size should be under 2 MB); 2) one figure for Question 1.2.(c) and two figures for Question 2.1.(c) in the .pdf format; and 3) two Python files classifier.py and ldaqda.py that contain your code. All these files should be uploaded to Quercus.

1 Naïve Bayes Classifier for Spam Filtering

1. (a) Write down the estimators for p_d and q_d as functions of the training data $\{\mathbf{x}_n, y_n\}, n = 1, 2, \dots, N$ using the technique of “Laplace smoothing”. (1 pt)
- $\rightarrow \text{email } \{1 \dots N\}$
 $\hookrightarrow \{0, 1\}$

Let x_{nd} represent: # times word w_d appears in email n

$$p_d = P(w_d | y_n = 1) = \frac{\sum_{n: y_n=1} x_{nd} + \alpha}{\sum_{n: y_n=1} \sum_{d=1}^D x_{nd} + \alpha D}$$

$$q_d = P(w_d | y_n = 0) = \frac{\sum_{n: y_n=0} x_{nd} + 1}{\sum_{n: y_n=0} \sum_{d=1}^D x_{nk} + 2D}$$

α : smooth parameter

- (b) Complete function learn_distributions in python file classifier.py based on the expressions you derived in part (a). (1 pt)
2. (a) Write down the MAP rule to decide whether $y = 1$ or $y = 0$ based on its feature vector $\mathbf{x}$ for a new email $\{\mathbf{x}, y\}$. The d -th entry of $\mathbf{x}$ is denoted by x_d . Please incorporate p_d and q_d in your expression. Please assume that $\pi = 0.5$. (1 pt)
- $P(y=1) = 0.5$

$$P(y|\mathbf{x}) \propto P(\mathbf{x}|y)P(y)$$

email is spam if: $\prod_{d=1}^D p_d^{x_d} \pi > \prod_{d=1}^D q_d^{x_d} (1-\pi)$

$$\xRightarrow{\log} \sum_{d=1}^D x_d \log p_d > \sum_{d=1}^D x_d \log q_d$$

- (b) Complete function `classify_new_email` in `classifier.py`, and test the classifier on the testing set. The number of Type 1 errors is 3, and the number of Type 2 errors is 4. (1 pt)
- (c) Write down the modified decision rule in the classifier such that these two types of error can be traded off. Please introduce a new parameter to achieve such a trade-off. (0.5 pt)

$$\begin{cases} \log P(y=1|x) - \log P(y=0|x) > T : \text{spam} \\ \text{o.w.} : \text{ham} \end{cases}$$

to decrease Type 2 error, we can turn down T

Write your code in file `classifier.py` to implement your modified decision rule. Test it on the testing set and plot a figure to show the trade-off between Type 1 error and Type 2 error. In the figure, the x -axis should be the number of Type 1 errors and the y -axis should be the number of Type 2 errors. Plot at least 10 points corresponding to different pairs of these two types of error in your figure. The two end points of the plot should be: 1) the point with zero Type 1 error; and 2) the point with zero Type 2 error. Please save the figure with name **nbc.pdf**. (1 pt)

3. Why do we need Laplace smoothing? Briefly explain what would go wrong if we do use the maximum likelihood estimators in the training process. (0.5 pt)

MLE: $p_d = \frac{\text{count of } w_d \text{ in spam}}{\text{total spam word}}$
 if a spam word never appear in training emails: $p_d = 0$. But that not mean that word is not a spam word, after adding Laplace smoothing: $p_d = \frac{\text{count} + 1}{\text{total spam words} + D}$
 No prob is forever zero, makes model becomes robust.